UE26MA141A · ENGINEERING MATHEMATICS – I · SHORT NOTES

Unit 1 — Partial Differentiation

Last-minute revision: definitions, formulas, standard results, worked answers and exam traps — built from the PESU Unit 1 slides (classes 1–11).

1. Limits, continuity & differentiability
2. Partial derivatives & geometric meaning
3. Total derivative, differentials & errors
4. Chain rule & composite functions
5. Implicit functions
6. Homogeneous functions & Euler's theorem
7. Taylor's & Maclaurin's series (2 variables)
8. Maxima, minima & saddle points
9. Lagrange's undetermined multipliers
10. One-page formula sheet

1 Limits, Continuity & Differentiability

A function of two variables $z = f(x, y)$ assigns one real number to each point (x, y) in its domain D. Examples: area $A = lb$, cylinder volume $V = \pi r^2 h$, potential $V = f(x, y, z, t)$.

▸ Limit

$\lim_{(x,y)\to(a,b)} f(x, y) = L$ means $f \to L$ along **every** path to (a, b). There are infinitely many paths.

- **Limit does not exist (DNE):** find two paths that give different values. Common paths: $y = 0$, $x = 0$, $y = mx$, $y = mx^2$.
- **To show a limit exists** at the origin, use polar form: $x = r\cos\theta$, $y = r\sin\theta$. If the result $\to L$ as $r \to 0$ for every θ, the limit is L.
- Polynomials: just substitute. E.g. $\lim_{(2,1)}(x^2 - 2xy + 3y^2 - 4x + 3y - 6) = 4 - 4 + 3 - 8 + 3 - 6 = \mathbf{-8}$.

Standard examples

(i) $\frac{2xy}{3x^2 + y^2}$ at (0,0): along $y = mx \to \frac{2m}{3 + m^2}$, which depends on m ($y = 0$ gives 0, $y = x$ gives ½) ⇒ **limit DNE ⇒ not continuous**.

(ii) $\frac{3x^2y}{x^2 + y^2}$ at (0,0): polar gives $3r\cos^2\theta\sin\theta \to 0$ ⇒ **limit = 0 exists**, but $f(0,0)$ is not defined ⇒ not continuous (the discontinuity is removable).

▸ Continuity

f is continuous at (a, b) if: (1) $f(a, b)$ is defined, (2) the limit exists, and (3) $\lim_{(x,y)\to(a,b)} f = f(a, b)$. Geometrically, the surface has no hole or tear at that point.

TRAP

"Not continuous" does **not** mean "the limit does not exist". The limit may exist while $f(a, b)$ is undefined or different (example ii).

▸ Differentiability

f is differentiable at (a, b) if $\Delta f = f_x h + f_y k + \varepsilon_1 h + \varepsilon_2 k$, where $\varepsilon_1, \varepsilon_2 \to 0$ as $(h, k) \to (0, 0)$.

- **Sufficient condition:** f_x and f_y exist and are continuous near (a, b).
- Differentiable ⇒ continuous. The converse is false.
- Partial derivatives can exist at a point where f is not continuous. Example: $xy/(x^2 + y^2)$ has $f_x(0,0) = f_y(0,0) = 0$, but it is discontinuous at the origin.

2 Partial Derivatives

A partial derivative is the derivative with respect to one variable while all the others are held **constant**.

$$\frac{\partial z}{\partial x} = z_x = f_x = \lim_{h\to 0} \frac{f(x+h,y)-f(x,y)}{h} \qquad \frac{\partial z}{\partial y} = z_y = f_y = \lim_{h\to 0} \frac{f(x,y+h)-f(x,y)}{h}$$

▸ Higher order

Notation	Meaning
$f_{xx} = \frac{\partial^2 f}{\partial x^2}$, $f_{yy} = \frac{\partial^2 f}{\partial y^2}$	Differentiate twice with respect to the same variable.
$f_{xy} = \frac{\partial^2 f}{\partial y\,\partial x} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right)$	Mixed partial: first x, then y. Subscripts read left → right; ∂-notation reads right → left.

- **Clairaut / Schwarz theorem:** $f_{xy} = f_{yx}$ when the mixed partials are continuous, so the order does not matter.
- A function of 2 variables has 2 first-order, 4 second-order and 2^n n-th order partial derivatives. A function of m variables has m^n of order n.

▸ Geometric interpretation

- $f_x(x_0, y_0)$ is the slope of the tangent to the curve cut from the surface $z = f(x, y)$ by the plane $y = y_0$ (the x-direction). f_y is the slope along the plane $x = x_0$.
- $f_{xx} > 0$: the surface is concave up in the x-direction. $f_{yy} > 0$: concave up in the y-direction.
- f_{xy} measures how the x-slope changes as we move in the y-direction.
- Tangent plane at P: $z - z_0 = f_x(x - x_0) + f_y(y - y_0)$.

▸ Frequently asked standard results

Given	Result
$r^2 = x^2 + y^2 + z^2$	$r_x = x/r,\ r_y = y/r,\ r_z = z/r$
$u = f(r), r = \sqrt{(x^2 + y^2 + z^2)}$	$u_x = f'(r)x/r;\ \ u_{xx} + u_{yy} + u_{zz} = f''(r) + \frac{2}{r}f'(r)$
$u = r^n$ (3-D)	$\nabla^2 u = n(n+1)r^{n-2}$; $u = 1/r$ satisfies Laplace's equation
$u = \log(x^2 + y^2),\ \ u = \tan^{-1}(y/x)$	Both satisfy $u_{xx} + u_{yy} = 0$ (harmonic)
$x = r\cos\theta, y = r\sin\theta$	$r_x = \cos\theta = x/r, r_y = \sin\theta, \theta_x = -y/r^2, \theta_y = x/r^2$

EXAM TIP
For "verify $u_{xy} = u_{yx}$" questions, compute both orders fully and show they are equal. For "show u satisfies Laplace", compute u_{xx} and u_{yy} separately, then add.

3 Total Derivative, Differentials & Errors

TOTAL DERIVATIVE (TREE RULE)

If $z = f(x, y)$ with $x = x(t), y = y(t)$: $\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$ (add a term for each extra variable).

TOTAL DIFFERENTIAL

$dz = f_x\,dx + f_y\,dy \quad \Rightarrow \quad f(a+h, b+k) \approx f(a,b) + h\,f_x(a,b) + k\,f_y(a,b)$

Example — direct
$z = x^2y^3, x = t^2, y = 2t^3$: $z_x = 2xy^3 = 16t^{11}, z_y = 3x^2y^2 = 12t^{10}$.
$\frac{dz}{dt} = 16t^{11}(2t) + 12t^{10}(6t^2) = \mathbf{104\,t^{12}}$. (Check: $z = 8t^{13}$.)

Example — approximation
$f(2, -3) = 6, f_x = 1.3, f_y = -0.6$. Estimate $f(2.1, -3.03)$:
$dz = 1.3(0.1) + (-0.6)(-0.03) = 0.148 \Rightarrow f \approx \mathbf{6.148}$.

▸ Errors & sensitivity

- Absolute error ≈ dz; relative error = dz/z; percentage error = $100\cdot dz/z$.
- If $u = k\,x^a\,y^b\,z^c$, take logs: $\frac{du}{u} = a\frac{dx}{x} + b\frac{dy}{y} + c\frac{dz}{z}$, so percentage errors add with the powers as weights.
- **Cylinder tank** ($h = 10, r = 2$): $dV = 2\pi rh\,dr + \pi r^2 dh = 40\pi\,dr + 4\pi\,dh$. The volume is about **10× more sensitive to the radius** than to the height.

▸ Rates-of-change problems (from the slides)

Problem	Key step → answer
Triangle, area constant; $x' = 3, y' = -2$ cm/s; $x = 20, y = 30, \theta = \pi/6$. Find θ'.	$A = \tfrac{1}{2}xy\sin\theta \Rightarrow 0 = \tfrac{1}{2}[y\sin\theta\cdot 3 + x\sin\theta\cdot(-2) + xy\cos\theta\,\theta']$ ⇒ $\theta' = -25/(300\sqrt{3}) \approx$ **−0.0481 rad/s** (θ decreasing)
Triangle: base 10 (+2 cm/s), height 6 (+1 cm/s). Rate of change of area?	$A' = \tfrac{1}{2}(b\,h' + h\,b') = \tfrac{1}{2}(10 + 12) =$ **11 cm²/s**
Cone, $r = h$, $V' = 9$ cm³/s. Find h' at $h = 6$.	$V = \pi h^3/3 \Rightarrow 9 = \pi(36)h' \Rightarrow h' =$ **$1/(4\pi) \approx 0.0796$ cm/s**
Point on $x^2/16 - y^2/9 = z$ and $x^2 + y^2 = 5$; $x' = 0.2$. Find z' at $x = 2$.	$dy/dx = -x/y$ (from the cylinder); use $z' = z_x x' + z_y y'$ with $y = \pm 1$

4 Chain Rule & Composite Functions

If $z = f(x, y), x = g(u, v), y = h(u, v)$:

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial u} \qquad \frac{\partial z}{\partial v} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial v}$$

Draw the **tree diagram**: multiply along each branch and add over all branches that end at the variable.

▸ Classic "prove that" results — introduce new variables *r*, *s*, *t*

Given	Prove
$v = f(2x - 3y, 3y - 4z, 4z - 2x)$	$6v_x + 4v_y + 3v_z = 0$ ($v_x = 2v_r - 2v_t, v_y = -3v_r + 3v_s, v_z = -4v_s + 4v_t$)
$u = f(x - y, y - z, z - x)$	$u_x + u_y + u_z = 0$
$u = f(x/y, y/z, z/x)$	$x\,u_x + y\,u_y + z\,u_z = 0$
$z = f(r, \theta), r = ts - t^2, \theta = \sqrt{(s^2 + t^2)}$	$z_s = f_r t + f_\theta s/\sqrt{(s^2 + t^2)};\ z_t = f_r(s - 2t) + f_\theta t/\sqrt{(s^2 + t^2)}$
$u = f(x, y), x = e^{st}, y = s^2 + t^2$	$u_s = f_x te^{st} + f_y 2s;\ \ u_t = f_x se^{st} + f_y 2t$
$u = f(r, \theta), r = \sqrt{(x^2 + y^2)}, \theta = \tan^{-1}(y/x)$	$u_x = f_r x/r - f_\theta y/r^2;\ \ u_y = f_r y/r + f_\theta x/r^2$

▸ Polar substitution results ($x = r\cos\theta$, $y = r\sin\theta$)

- $u = f(x^2 + y^2)$ (for example $\sin(x^2 + y^2)$ or $e^{x^2+y^2}$) ⇒ $\frac{\partial u}{\partial \theta} =$ **0**, because u depends only on r.
- $u = \tan^{-1}(y/x) \Rightarrow u_\theta = u_x(-r\sin\theta) + u_y(r\cos\theta) = r^2(\sin^2\theta + \cos^2\theta)/r^2 =$ **1**.
- $u = x^2y + yz^2$, with $x = r\cos\theta, y = r\sin\theta, z = \theta \Rightarrow u_\theta = 2xy(-r\sin\theta) + (x^2 + z^2)(r\cos\theta) + 2yz$.
- Identity: $(u_x)^2 + (u_y)^2 = (u_r)^2 + \frac{1}{r^2}(u_\theta)^2$.

Example — one-variable chain

$u = e^{xy}\cdot$(stuff) problems: always write $\frac{du}{dt} = u_x x' + u_y y'$, then substitute $x(t)$ and $y(t)$ **at the end**. E.g. $u = e^{xy} + \ln(y^2 + 1)$, $x = \tan t, y = \cos t$: $\frac{du}{dt} = y\,e^{xy}\sec^2 t + \left(x\,e^{xy} + \frac{2y}{y^2 + 1}\right)(-\sin t)$.

5 Implicit Functions

TWO VARIABLES: F(X, Y) = 0

$$\frac{dy}{dx} = -\frac{f_x}{f_y},\ f_y \neq 0$$

$$\frac{d^2y}{dx^2} = -\frac{f_{xx}f_y^{\,2} - 2f_{xy}f_xf_y + f_{yy}f_x^{\,2}}{f_y^{\,3}}$$

THREE VARIABLES: F(X, Y, Z) = 0

$$\frac{\partial z}{\partial x} = -\frac{f_x}{f_z},\quad \frac{\partial z}{\partial y} = -\frac{f_y}{f_z}$$

$$\frac{\partial y}{\partial x} = -\frac{f_x}{f_y}$$ (holding z constant)

Reason: on a level curve $f(x, y) = c$, $df = f_x dx + f_y dy = 0$.

Problem	Answer
$x^3y + 3xy^2 = 3$	$dy/dx = -3y(x^2 + y)/[x(x^2 + 6y)]$
$x\,e^y + y\,e^x = 0$: tangent at the origin	$dy/dx = -(e^y + y\,e^x)/(x\,e^y + e^x) = -1$ at $(0,0) \Rightarrow$ tangent $\boldsymbol{x + y = 0}$
$(\cos x)^y = (\sin y)^x$	$dy/dx = \frac{y\tan x + \log\sin y}{\log\cos x - x\cot y}$ (take logs: $F = y\log\cos x - x\log\sin y$, then $-F_x/F_y$)
$z^3 + xy - y^2z = 6$ at $(0, 1, 2)$	$z_x = -y/(3z^2 - y^2) = \mathbf{-1/11}$; $y_x = -y/(x - 2yz) = \mathbf{1/4}$
$x^2 + xy + y^2 = 7$	$dy/dx = -(2x + y)/(x + 2y)$
$x^3 + z^2 + y\,e^{xz} + z\cos y = 0$	$z_x = -(3x^2 + yz\,e^{xz})/(2z + xy\,e^{xz} + \cos y) \Rightarrow 0$ at $(0,0,0)$

6 Homogeneous Functions & Euler's Theorem

DEFINITION

$u = f(x, y)$ is homogeneous of degree n if $\boldsymbol{f(\lambda x, \lambda y) = \lambda^n f(x, y)}$, or equivalently $u = x^n\,g(y/x) = y^n\,g(x/y)$. For three variables: $u = x^n\,g(y/x, z/x)$. The degree n can be positive, negative, zero or a fraction.

EULER'S THEOREM

$$x\,u_x + y\,u_y = n\,u$$

$$x\,u_x + y\,u_y + z\,u_z = n\,u \quad (3 \text{ variables})$$

COROLLARY (SECOND ORDER)

$$x^2u_{xx} + 2xy\,u_{xy} + y^2u_{yy} = n(n-1)\,u$$

MODIFIED EULER — F(U) HOMOGENEOUS, U NOT

If $f(u)$ is homogeneous of degree n:

$$x\,u_x + y\,u_y = n\,\frac{f(u)}{f'(u)} = g(u)$$

$$x^2u_{xx} + 2xy\,u_{xy} + y^2u_{yy} = g(u)[g'(u) - 1]$$

- A constant has degree 0; linear functions have degree 1. In thermodynamics, extensive variables have degree 1 and intensive variables degree 0. Euler's theorem is also used in economics (production functions).
- **Degree checks from the slides:** $(x^4 + 2x^2y^2 + y^4)/(x^2 + y^2)^2 \to 0$; $\sin^{-1}(y/x)/\sqrt{(x^2 + y^2)} \to -1$; $(x^2y + xy^2)/(x + y)^3 \to 0$; $(e^x - e^y)/(x - y) \to$ **not homogeneous**.

▸ Euler's theorem — standard problems (memorise the pattern)

Given u	Homogeneous part & degree	$x u_x + y u_y$ (+ $z u_z$)	$x^2u_{xx} + 2xy\,u_{xy} + y^2u_{yy}$
$u = (x-y)/(x^2+y^2)$	u itself, $n = -1$	$-u$	$(-1)(-2)u = 2u$
$u = \sin^{-1}[(x+2y+3z)/\sqrt{(x^8+y^8+z^8)}]$	$\sin u$, $n = 1-4 = -3$	$-3\tan u \Rightarrow$ sum + $3\tan u = 0$	—
$u = \sin^{-1}[(x+y)/(\sqrt{x}+\sqrt{y})]$	$\sin u$, $n = ½$	$½\tan u$	$½\tan u\,(½\sec^2 u - 1) = -\frac{\sin u \cos 2u}{4\cos^3 u}$
$u = \log[(x^2+y^2)/(x+y)]$	e^u, $n = 1$	1	$1\cdot(0-1) = -1$
$u = \exp[x^3y^3/(x^2+y^2)]$	$\log u$, $n = 4$	$4u \log u$	—
$u = \tan^{-1}[(x^3+y^3)/(x-y)]$	$\tan u$, $n = 2$	$2\sin u \cos u = \sin 2u$	$\sin 2u(2\cos 2u - 1) = \sin 4u - \sin 2u$
$z = (x+y)/(\sqrt{x}+\sqrt{y})$	z, $n = ½$	$½\,z$ (verify directly)	$-¼\,z$

METHOD

1) Check whether u is homogeneous using $u(\lambda x, \lambda y)$. 2) If not, look for $f(u)$ ($\sin u$, $\tan u$, e^u, $\log u$) that is homogeneous. 3) Apply Euler to $f(u)$: $f'(u)(x\,u_x + y\,u_y) = n\,f(u)$. 4) For the second-order result use $g(u)[g'(u) - 1]$. Write $g'(u)$, **not** $g(u)$, inside the bracket.

7 Taylor's & Maclaurin's Series

▸ One variable (review)

$f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \ldots$; with $a = 0$ this is Maclaurin's series.

$e^x = 1 + x + x^2/2! + x^3/3! + \ldots$	$\sin x = x - x^3/3! + x^5/5! - \ldots$	$\cos x = 1 - x^2/2! + x^4/4! - \ldots$
$\log(1+x) = x - x^2/2 + x^3/3 - \ldots\ (\lvert x\rvert < 1)$	$\sinh x = x + x^3/3! + x^5/5! + \ldots$	$\cosh x = 1 + x^2/2! + x^4/4! + \ldots$
$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \ldots$	$\tan^{-1}x = x - x^3/3 + x^5/5 - \ldots$	$\sin^{-1}x = x + \frac{1}{2}\frac{x^3}{3} + \frac{1\cdot3}{2\cdot4}\frac{x^5}{5} + \ldots$

▸ Two variables — about (a, b), with $h = x - a$, $k = y - b$

$$f(x,y) = f(a,b) + \frac{1}{1!}[h f_x + k f_y] + \frac{1}{2!}[h^2 f_{xx} + 2hk f_{xy} + k^2 f_{yy}] + \frac{1}{3!}[h^3 f_{xxx} + 3h^2k f_{xxy} + 3hk^2 f_{xyy} + k^3 f_{yyy}] + \ldots$$

All derivatives are evaluated at (a, b). Compact form: $f(a+h, b+k) = \Sigma\,\frac{1}{n!}(h\partial_x + k\partial_y)^n f(a,b)$, with binomial coefficients 1-2-1 and 1-3-3-1.

Maclaurin ($a = b = 0$): $f(x,y) = f(0,0) + [x f_x + y f_y] + \frac{1}{2!}[x^2 f_{xx} + 2xy f_{xy} + y^2 f_{yy}] + \ldots$

WORKING RULE

Make a table: $f, f_x, f_y, f_{xx}, f_{xy}, f_{yy}$ (and the third-order derivatives if asked), each with its value at (a, b), then substitute. "Expand in powers of $(x-1)$ and $(y+2)$" means $a = 1$, $b = -2$.

▸ Worked expansions (final answers)

Function & point	Values at the point	Expansion
$e^x \cos y$ about $(1, \pi/4)$	$f = f_x = f_{xx} = e/\sqrt{2}$; $f_y = f_{xy} = f_{yy} = -e/\sqrt{2}$	$\frac{e}{\sqrt{2}}[1 + h - k + \frac{1}{2}(h^2 - 2hk - k^2)]$ with $h = x - 1, k = y - \pi/4$
$\log(2x + y + 1)$ about $(0,0)$	0; 2, 1; −4, −2, −1	$2x + y - \frac{1}{2}(4x^2 + 4xy + y^2) = (2x + y) - \frac{1}{2}(2x + y)^2$
$\tan^{-1}(xy)$ about $(1, -1)$; estimate $f(0.9, -1.2)$	$f = -\pi/4; f_x = -\frac{1}{2}, f_y = \frac{1}{2}; f_{xx} = \frac{1}{2}, f_{xy} = 0, f_{yy} = \frac{1}{2}$	$-\pi/4 - \frac{1}{2}h + \frac{1}{2}k + \frac{1}{4}h^2 + \frac{1}{4}k^2$; with $h = -0.1, k = -0.2$: ≈ **−0.823**
e^{x-y} about $(0,0)$	all derivatives ±1	$1 + (x - y) + (x - y)^2/2! + (x - y)^3/3! + \ldots$
$e^x \log(1 + y)$, first six terms	$f_y = 1, f_{xy} = 1, f_{yy} = -1, f_{xxy} = 1, f_{xyy} = -1, f_{yyy} = 2$	$y + xy - y^2/2 + x^2y/2 - xy^2/2 + y^3/3$
$\sin(xy)$ about $(1, \pi/2)$	$f = 1; f_x = f_y = 0; f_{xx} = -\pi^2/4, f_{xy} = -\pi/2, f_{yy} = -1$	$1 - \frac{\pi^2}{8}h^2 - \frac{\pi}{2}hk - \frac{1}{2}k^2$ $(h = x - 1, k = y - \pi/2)$
$xy^2 + y\cos(x - y)$ about $(1,1)$	$f = 2; 1, 3; f_{xx} = -1, f_{xy} = 3, f_{yy} = 1$	$2 + h + 3k + \frac{1}{2}[-h^2 + 6hk + k^2]$

Uses: approximating values, polynomial models, optimisation (quadratic approximations), error and variance estimates; also why $\sin\theta \approx \theta$ for a pendulum.

8 Maxima & Minima (two variables)

$f(a, b)$ is a **relative maximum** if $f(a, b) > f(a + h, b + k)$ for all small h, k ($\Delta = f(a + h, b + k) - f(a, b) < 0$), and a **relative minimum** if $\Delta > 0$. Maxima and minima together are called **extreme values**. A **saddle point** is a stationary point that is neither.

SECOND-DERIVATIVE TEST

Stationary points: solve $f_x = 0$ and $f_y = 0$. Let $r = f_{xx}, s = f_{xy}, t = f_{yy}$. Then $rt - s^2$ = det of the Hessian $[[f_{xx}, f_{xy}], [f_{yx}, f_{yy}]]$.

$rt - s^2$	r	Nature of (a, b)
> 0	< 0	**Maximum**
> 0	> 0	**Minimum**
< 0	any	**Saddle point** (not an extremum)
= 0	—	Doubtful: the test fails, investigate further

MEMORY HOOK

"**r** negative ⇒ **max**, like $f'' < 0$ in one variable." Always check the sign of $rt - s^2$ **first**.

▸ Solved maxima/minima problems

Function	Stationary points	Result
$x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$	$f_y = 6y(x-5) \Rightarrow y = 0$ or $x = 5 \Rightarrow (6,0), (4,0), (5,\pm1)$	$rt - s^2 = 36[(x-5)^2 - y^2]$. (6,0): **min = 108**; (4,0): **max = 112**; (5,±1): saddle
$x^4 + y^4 - x^2 - y^2 + 1$	$x, y \in \{0, \pm1/\sqrt{2}\}$	(0,0): **max = 1**; $(\pm1/\sqrt{2}, \pm1/\sqrt{2})$: **min = ½**; $(\pm1/\sqrt{2}, 0)$, $(0, \pm1/\sqrt{2})$: saddle
$\sin x + \sin y + \sin(x+y)$	$\cos x = \cos y \Rightarrow x = y$, $\cos 2x = -\cos x \Rightarrow x = \pi/3$	At $(\pi/3, \pi/3)$: $r = -\sqrt{3} < 0$, $rt - s^2 = 9/4 > 0 \Rightarrow$ **max = $3\sqrt{3}/2$**
$T = 400xyz^2$ on $x^2 + y^2 + z^2 = 1$	Eliminate z^2: $F = 400xy(1 - x^2 - y^2)$; points (0,0), $(\pm½, \pm½)$	(0,0): saddle; max at $(½, ½)$, $(-½, -½)$: $\boldsymbol{T_{max} = 50}$
$T = kxyz^2$ on $x^2 + y^2 + z^2 = a^2$	(0,0), $(\pm a/2, \pm a/2)$	$\boldsymbol{T_{max} = ka^4/8}$ at $x = y = a/2$ (and $-a/2$)
Open box, volume V, least metal: $S = xy + 2V/x + 2V/y$	$S_x = 0, S_y = 0 \Rightarrow x = y = (2V)^{1/3}$	$z = x/2$ (base square, height half the side). For $V = 10$: $x = y = \sqrt[3]{20} \approx 2.71$, $z \approx 1.36$

9 Lagrange's Method of Undetermined Multipliers

Use this method to find the extremum of $f(x, y, z)$ **subject to a constraint** $\varphi(x, y, z) = 0$.

WORKING PROCEDURE

1. Form $\boldsymbol{F = f + \lambda\varphi}$ (λ is the Lagrange multiplier).
2. Solve $F_x = 0, F_y = 0, F_z = 0$ together with $\varphi = 0$ (4 equations in x, y, z, λ).
3. **Trick:** write λ from each equation and equate, for example $\lambda = -yz/(2x) = -xz/(2y) \Rightarrow x^2 = y^2$. Then substitute into φ.

Advantages: stationary values are found without solving for the variables explicitly at every step, and the method extends to many variables and constraints.

Disadvantages: it does not tell whether a point is a max or a min (use physical reasoning), and the conditions $F_x = F_y = F_z = 0$ are only necessary conditions.

▸ Standard results

Problem	Setup → Answer
Rectangular solid of max volume inscribed in a sphere $x^2 + y^2 + z^2 = a^2$	$V = 8xyz$, $\varphi = x^2 + y^2 + z^2 - a^2$. $8yz + 2\lambda x = 0$, etc. $\Rightarrow x^2 = y^2 = z^2 \Rightarrow$ **cube**, side $2a/\sqrt{3}$, $V_{max} = 8a^3/(3\sqrt{3})$
Points on $x^2 - z^2 = 1$ closest to the origin	$f = x^2 + y^2 + z^2$: $2x(1 + \lambda) = 0, 2y = 0, 2z(1 - \lambda) = 0$. $x \neq 0 \Rightarrow \lambda = -1 \Rightarrow z = 0 \Rightarrow$ **(±1, 0, 0)**, distance 1
Open box, surface $xy + 2yz + 2zx = 432$, max volume	$x = y = 2z \Rightarrow 3x^2 = 432 \Rightarrow$ **12 × 12 × 6 cm**, $V = 864$ cm^3
Divide 24 into $x + y + z$ with xy^2z^3 maximum	$\lambda = -y^2z^3 = -2xyz^3 = -3xy^2z^2 \Rightarrow y = 2x, z = 3x \Rightarrow 6x = 24 \Rightarrow$ **4, 8, 12**
Sum fixed ($x + y + z = k$), product xyz max	**Equal parts**: $x = y = z = k/3$. General rule: $x^py^qz^r$ is max when $x : y : z = p : q : r$

★ One-Page Formula Sheet — Read This Before the Exam

LIMITS & CONTINUITY

Two paths give different limits ⇒ DNE. Polar form $r \to 0$ to prove existence.
Continuous ⇔ lim = $f(a, b)$. Differentiable ⇒ continuous. Partial derivatives existing does not imply continuity.

PARTIAL DERIVATIVES

$f_{xy} = f_{yx}$ (continuous mixed partials).
$r^2 = x^2 + y^2 + z^2 \Rightarrow r_x = x/r$.
$u = f(r)$: $\nabla^2 u = f'' + (2/r)f'$ (3-D), $f'' + (1/r)f'$ (2-D).

TOTAL DERIVATIVE / DIFFERENTIAL

$dz/dt = z_x x' + z_y y'$
$dz = f_x dx + f_y dy$; $f(a + h, b + k) \approx f + hf_x + kf_y$
$u = x^a y^b \Rightarrow$ % error in $u = a$(% in x) + b(% in y)

CHAIN RULE

$z_u = z_x x_u + z_y y_u$, $z_v = z_x x_v + z_y y_v$
$f(x - y, y - z, z - x) \Rightarrow u_x + u_y + u_z = 0$
$f(x/y, y/z, z/x) \Rightarrow xu_x + yu_y + zu_z = 0$

IMPLICIT

$dy/dx = -f_x/f_y$; $z_x = -f_x/f_z$, $z_y = -f_y/f_z$

EULER'S THEOREM

Degree n: $f(\lambda x, \lambda y) = \lambda^n f$
$xu_x + yu_y = nu$
$x^2 u_{xx} + 2xyu_{xy} + y^2 u_{yy} = n(n - 1)u$
If $f(u)$ is homogeneous: $g(u) = n\,f(u)/f'(u)$;
second order = $g(u)[g'(u) - 1]$
$\sin u \to n \tan u$; $\tan u \to (n/2) \sin 2u$; $e^u \to n$; $\log u \to n\,u \log u$

TAYLOR (TWO VARIABLES)

$$f = f + (hf_x + kf_y) + \frac{1}{2!}(h^2 f_{xx} + 2hkf_{xy} + k^2 f_{yy}) + \frac{1}{3!}(h^3 f_{xxx} + 3h^2 k f_{xxy} + 3hk^2 f_{xyy} + k^3 f_{yyy})$$

$h = x - a, k = y - b$; Maclaurin: $a = b = 0$

MAX / MIN

$f_x = f_y = 0$; $r = f_{xx}, s = f_{xy}, t = f_{yy}$
$rt - s^2 > 0, r < 0 \to$ max | $r > 0 \to$ min
$rt - s^2 < 0 \to$ saddle | $= 0 \to$ doubtful

LAGRANGE

$F = f + \lambda\varphi$; $F_x = F_y = F_z = 0$ and $\varphi = 0$.
Equate the expressions for λ. Sphere → cube; open box (surface fixed) → $x = y = 2z$.

▸ Common mistakes that cost marks

- Mixing up f_{xy} (first ∂x, then ∂y) with $\partial^2 f/\partial x \partial y$ (first ∂y). The answer is the same when the partials are continuous, but state it correctly.
- In implicit differentiation, forgetting the minus sign in $-f_x/f_y$.
- Applying Euler to u when only $f(u)$ is homogeneous. Always test the degree with λ first.
- Not converting units (inches → feet, degrees → radians) in rate and error problems.
- Max/min: forgetting to list **all** stationary points (factor completely, consider every case such as $y = 0$ or $x = 5$) and to state the max/min **value**, not just the point.
- Taylor: forgetting the factor 2 in $2hkf_{xy}$ and the 1/2! in front of the bracket.

Exam-hall checklist
✔ Write the formula first (it earns step marks) ✔ Draw a tree diagram for chain-rule questions ✔ Make a derivative table for Taylor and max/min ✔ Box the final answer with units.

Source: PESU UE26MA141A Unit 1 slides (classes 1–11) and lecture summaries. Some slide formulas were images, so the worked answers above were re-derived and checked.